Language Features under Weight Quantization

Activation Agreement and Intervention Effects

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The question

When a model's weights are rounded to fewer bits,

do the internal features that **explain** its behaviour

still work the same way?

We study language-selective features across 27 languages,
three quantizers, and two Gemma model sizes.

Why this matters

- Quantization makes LLMs cheaper to run, but can change behaviour
- Effects differ across languages – low-resource languages may suffer more
- Sparse autoencoders (SAEs) find features that fire for specific languages
- Prior work (Duan, 2026) measured *activation survival* under quantization

Our contribution: we go beyond activation survival and ask whether features still *do the same thing* when edited

Three things that come apart

Measure	What it tests	INT4 legacy
Activation survival ($R_1 > 0.9$)	Does the feature fire the same way?	13.2%
Functional effect (E)	Does removing it still hurt the language?	27/27
Effect identity (W_1)	Does it still do the same thing to output?	99.7%

The activation cutoff says 87% of features are "dead."
But 99.7% can still be identified from what they do.

Setup

Models

- Gemma-2-2B (main)
- Gemma-2-9B (confirmation)
- Llama-3.2-1B (cross-check)

Languages

27 from FLORES-200

9 high / 9 mid / 9 low resource

Quantizers

- Legacy RTN (from prior work)
- Matched RTN (same method, better implementation)
- GPTQ (calibrated rounding)

SAE

Gemma Scope JumpReLU, 16k features

One fixed layer per model size

What we measure

For each language, select 64 features that fire most selectively for it

1. **Zero-ablate** the group: set those 64 features to zero
2. Measure the increase in next-token loss (NLL)
3. Compare against **matched controls**: non-selective features with similar firing rates
4. Net harm E = group harm − control harm

Do the same at full precision and at each bit width.
The **retention ratio** $\rho = E^{(b)}/E^{\mathrm{FP}}$ tells us if the edit matters more or less after quantization.

Correction: the missing BOS token

Reproducing Duan (2026), we found a methodological issue:

	Without BOS	With BOS
FP perplexity	458.4	**16.1**
INT6 survival	51%	**97.5%**
INT7 survival	81%	**>99.9%**

One missing token at the start of the input made the model look 28× worse at prediction and made half the features look "dead" at INT6.

We corrected our protocol and retired our own first pipeline after finding separate flaws.

Features matter in all 27 languages

Removing a language's feature group hurts next-token prediction more than removing matched controls.

- **FP:** positive excess harm in 27/27 languages (median 0.262 nats/token)
- **INT6:** retention ratio ≈ 1.0 (features matter about as much)
- **INT4:** retention ratio 1.54 (legacy RTN) – the edit hurts *more*

Gradual suppression confirms dose–response in all 27 languages: as you turn features down, harm increases monotonically.

But features carry almost none of the damage

If features are hurt more at INT4, maybe fixing them fixes the model?

Rescue experiment: patch FP activations back into the quantized model.

What we restore	% of INT4 loss gap recovered
Language group (64 features)	2.6%
Full residual stream	39.3%

The amplified ratio does not mean the features carry the quantization damage. The compressed model is just more sensitive to any perturbation.

The direction-swap control

Is the INT4 similarity loss *specific* to each feature?

Control: ablate a *different* feature's direction at the same positions and strengths.

	Real feature	Swapped direction
Median W_1 at INT4	0.842	0.828

The gap is 0.014 – consistent at every bit width.

The similarity loss reflects how the *downstream computation* drifts under quantization for **any** perturbation, not anything specific to the feature.

GPTQ helps substantially

Three quantizers on Gemma-2-2B at INT4

	Legacy RTN	Matched RTN	GPTQ
Survival (%)	12.7	18.6	**30.5**
Mean loss (nats/tok)	0.590	0.385	**0.130**
Median ρ	1.54	1.36	**1.22**
Median W_1	0.839	0.875	**0.923**

- GPTQ roughly halves the INT4 amplification
- Matched RTN alone (better implementation, same method) closes about a third of the gap
- At INT6, all three quantizers preserve features well

Gemma-2-9B: confirmed

We ran the full study on a second model size with all three quantizers.

9B at INT4	Legacy	Matched	GPTQ
Survival (%)	22.3	32.9	**46.6**
Loss (nats/tok)	0.410	0.287	**0.088**
Amplification ρ	1.69	1.49	**1.17**

- GPTQ below legacy in **26 of 27** languages
- All frozen predictions (written before any 9B data existed) were met
- The pattern holds across model sizes within the Gemma family

Honest limits

Belebele null

Our only confirmatory test.
Removing a language's features does *not* change reading-comprehension accuracy.

FP effect: -0.0001 nats/item
$[-0.0035, +0.0031]$

The features matter for text prediction but not for this comprehension task.

Llama-3.2-1B: not consistent

Positive FP effect in only 21/27 languages (rule: 24).
INT4 ratio: 0.86 (no amplification).

But model family, SAE architecture, and tokenizer all change together, so we cannot isolate the cause.

Most FLORES analyses are exploratory (sentences reused). Only Belebele was held out.

What we claim

1. Language-selective SAE features contribute to text prediction in **all 27 languages** on both Gemma sizes
2. **Activation survival understates preservation**: 99.7% of features retain their identity even when R_1 says they are dead
3. The INT4 similarity loss is **generic**, not feature-specific (direction-swap control)
4. The features carry only **2.6%** of INT4 quantization damage (rescue study)
5. **GPTQ roughly halves** the INT4 amplification compared to RTN
6. The pattern **holds on 9B** but **not on Llama**

The Belebele null and rescue limit show where the features *don't* matter.

Corrections and integrity

- Found and reported the BOS error in prior work (reproduced their numbers first)
- Retired our own first pipeline after finding separate flaws
- Two 9B predictions were defective (reversed inequality, tautological criterion) – documented, not counted
- Frozen predictions written before data; code hashes locked; controls pre-specified
- Belebele null and Llama failure reported as results, not hidden

Resources:

Code: `github.com/TH4KUR/always-late`

Artifacts: `huggingface.co/datasets/karma-skz/always-late-artifacts`

Runs: `wandb.ai/karma-skz-/anlp-project`